

Beyond Weighted Interleaving: Bandwidth-Driven Memory Tiering with DAMON

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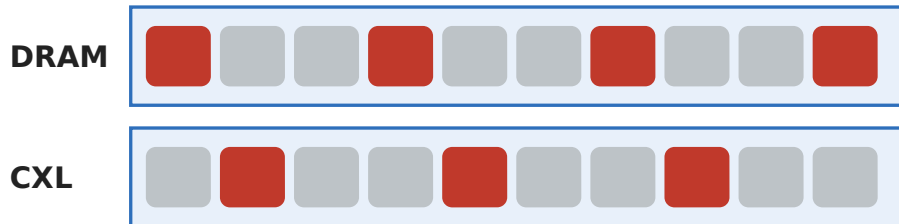
LINUX
PLUMBERS CONFERENCE

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Each Policy Solves Half

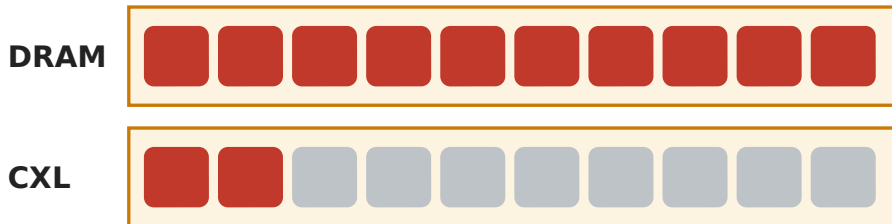
WEIGHTED INTERLEAVING · fixed split



✓ drives both memory paths at once

✗ blind to which pages are hot

HOTNESS TIERING · promote what is hot

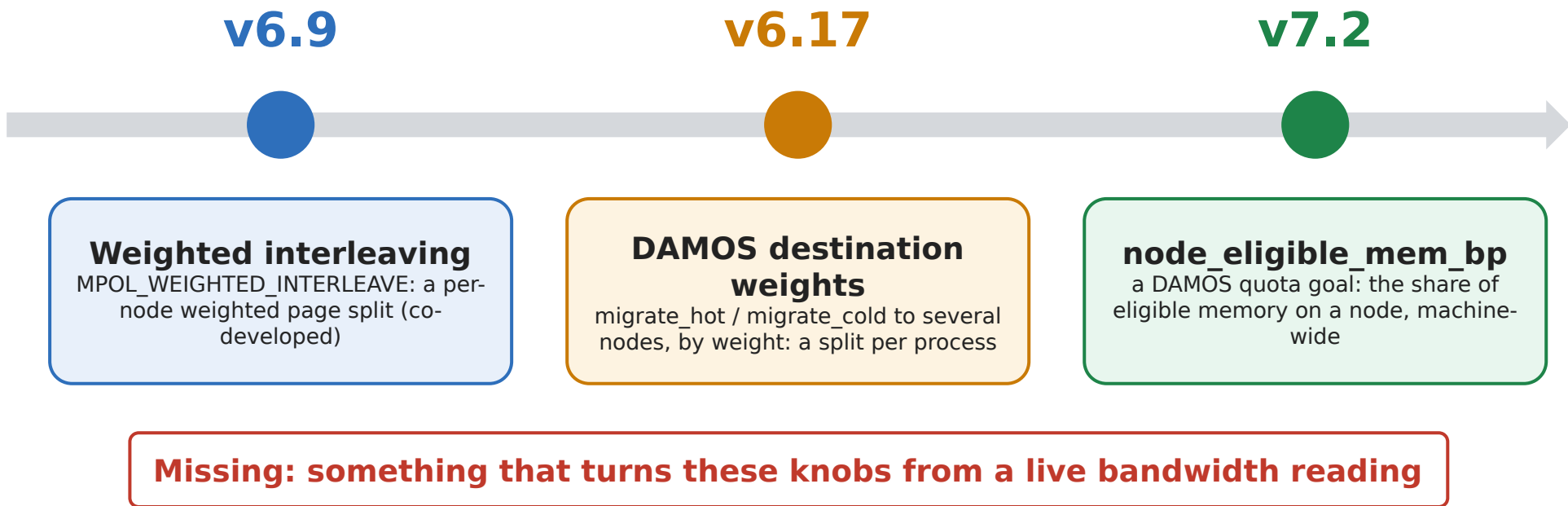


✓ right about heat

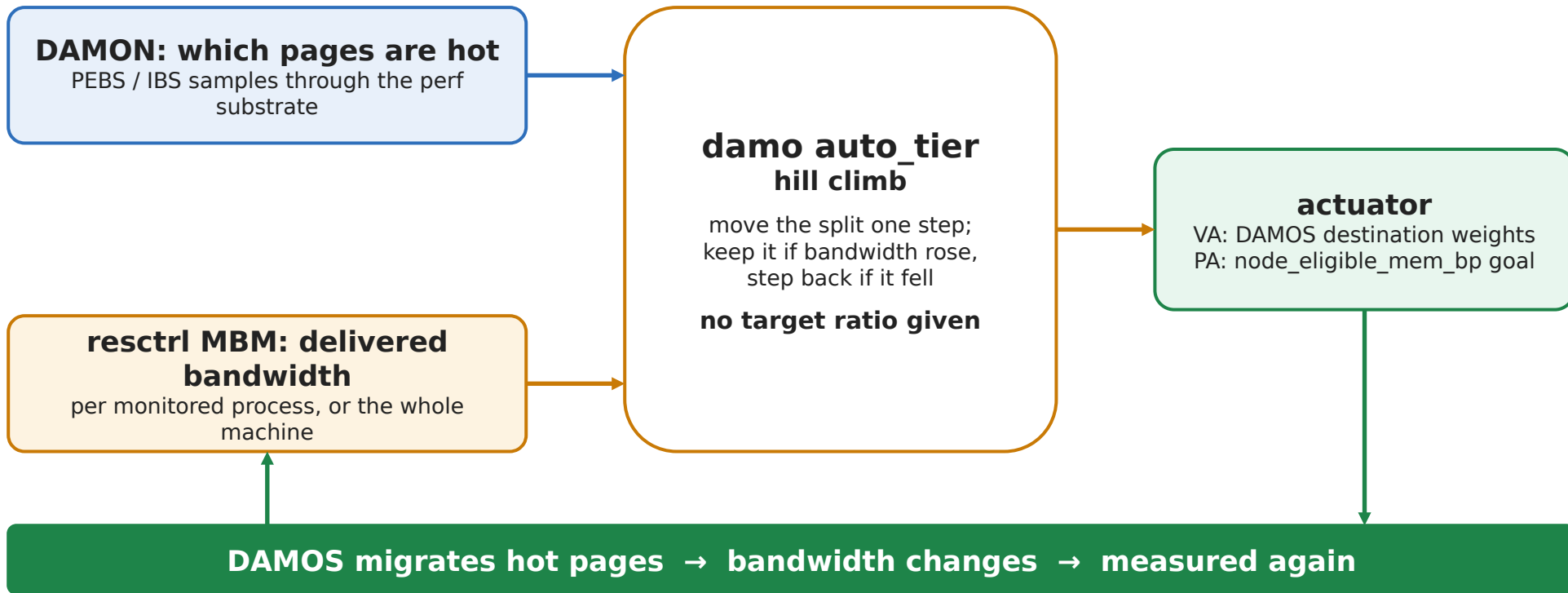
✗ keeps filling DRAM after it stops helping

Wanted: place by hotness, split by the bandwidth the machine actually delivers

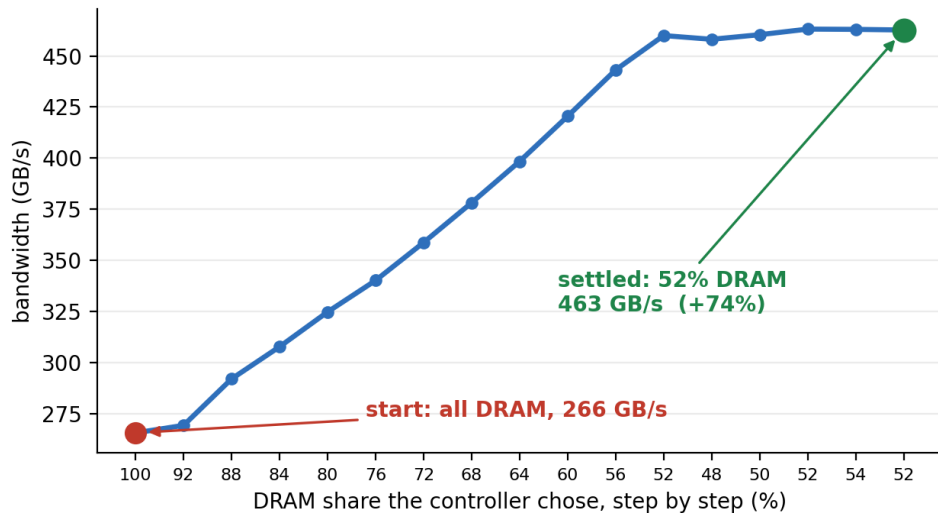
The Knobs Are Already Upstream



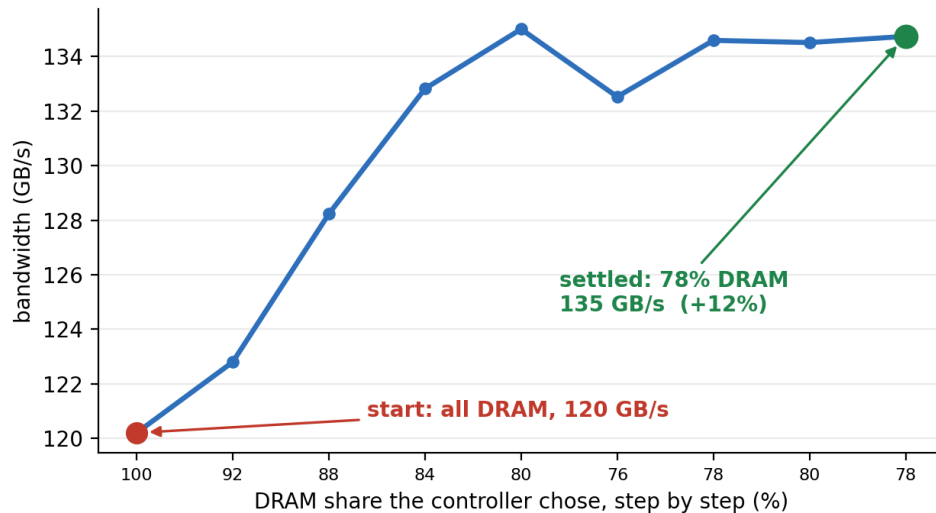
Closing the Loop



It Finds the Best Split on Its Own



Intel PEBS · virtual addresses · named processes
266 → 463 GB/s, settled at 52% DRAM



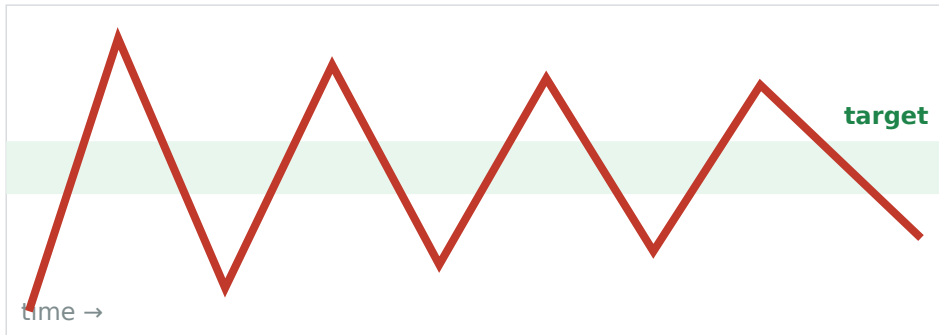
AMD IBS · physical addresses · whole machine
120 → 135 GB/s, settled at 78% DRAM

Both start all-DRAM, with no target and no model of the machine

Why Hardware Samples Settle

ACCESSED-BIT SCAN · a delayed view

pages move, but the scan reports it
several aggregation intervals later
so the controller corrects an error already fixed



✗ overshoots, then hunts around the target

Measured, accessed-bit scan

even with a cache workaround: in band only 15-22% of the time

HARDWARE SAMPLES · a current view

the next samples land where
the pages are now
so the goal is measured against real placement



✓ reaches the target and holds

Measured, IBS samples

the climb reached its peak in 4 steps (previous slide)

Traces are illustrative.

What Next: for Discussion

Upstream path

RFC v4 on damon/next; fold into milestone 2's report path with Akinobu

Where the controller lives

userspace damo today; which parts belong in DAMOS goal tuning?

More hotness sources

sampling PMUs feed the overflow path; trace PMUs such as Arm SPE via Kunwu's AUX backend

Observability

damos_node_eligible_mem_bp tracepoint, posted for mm-new

Kernel: [RFC v4 cover letter](#) · [tracepoint patch](#)

Userspace: [damo_auto_tier branch](#)